3. Is LLM self-consistency a usable signal for routing annotation work?

Supervisor: Soveatin Kuntur **Difficulty:** **Hard (14/25)**

Example of a research question	Does an LLM's self-consistency across reruns predict whether its label is actually correct — well enough to decide which items a model can annotate and which must go to a human?
Why it matters	Annotation is the expensive step in this field, and LLMs are increasingly used for it, usually in a single run with a single prompt. If a cheap rerun agreement predicted correctness, annotation budgets could be routed rationally; if it does not, a practice people already rely on is unsound.
Data — READY / TO COLLECT / NDA	Status: READY ● Gold labels: Polish split of SemEval-2023 Task 3 (23 persuasion techniques, paragraph level, aclanthology.org/2023.semeval-1.317) and MIPD manipulation-technique labels (github.com/ArkadiusDS/MIPD). Optional: Topic A's headline device labels, as a deliberately shallower task for contrast. ● Models: speakleash/Bielik-11B-v3.0-Instruct and CYFRAGOVPL/PLLuM-12B-chat as Polish open-weight models, plus one general multilingual open model — check each licence, several are CC BY-NC. ● Compute: reruns multiply cost, so budget for roughly 5 reruns x 3 prompt variants x 3 models over a fixed item sample rather than the full corpus.
What you will do	● Run each model 5 times per item across 3 prompt variants with fixed decoding, and compute per-item self-consistency alongside correctness against gold. ● Measure whether consistency predicts correctness — AUC per model and per label — and compare it against the obvious alternatives: token log-probability, verbalised confidence, and random routing. ● Build the routing curve: annotation quality and human hours saved as a function of how much of the data is left to the model, and name an operating point a real project could adopt.
What success looks like	● A consistency-versus-correctness result per model and per label, with confidence intervals, benchmarked against log-probability and verbalised-confidence baselines. ● A cost-quality routing curve with a stated operating point, or the clear negative finding that consistency carries no usable signal. ● Released prompts, logging code and raw outputs so the runs can be repeated.
Out of scope	Routing anything other than text-observable technique labels — attribution of a campaign to an actor or state is never automated here. Also out: generating synthetic disinformation and fine-tuning any model.
Starting point	Prior work: Rerun variance in LLM annotation has been reported before, so instability on its own is NOT the contribution. New this year: whether that variance is diagnostic of error, benchmarked against standard confidence signals, and evaluated on Polish models rather than English-centric ones.
Data & ethics	Prompts contain real manipulative content. Do not publish model completions verbatim where they read as fresh disinformation; report aggregate results and keep raw outputs in a private folder that the report describes.

Level & skills	Level: Hard (score 14/25, see the difficulty table below) Skills: Python, prompting and running open-weight LLMs (GPU), evaluation of confidence signals (AUC) and basic statistics.
Contact & availability	Soveatin Kuntur, M.Sc. Eng. — soveatin.kuntur.dokt@pw.edu.pl Availability: Wednesdays only, during NLP course hours.
Team slots	Max 2 teams: one on Polish open-weight models, one on multilingual models and the confidence baselines. Same items, same prompts, poolable results.

Difficulty dimensions

Dimension	Value (1–5)	Basis in this proposal
D1 Data work	2	Gold labels (SemEval-2023 Task 3, MIPD) used as released; fixed item sample.
D2 Compute and infrastructure	4	5 reruns × 3 prompt variants × 3 models (Bielik-11B, PLLuM-12B, one multilingual model).
D3 Technical depth	3	Prompted annotation, uncertainty signals and a routing curve.
D4 Methodological demands	2	2 of 6 elements: confidence intervals, ≥3 alternative signals compared (log-probability, verbalised confidence, random routing).
D5 External dependencies and risks	3	3 stated dependencies: licences (several CC BY-NC), manipulative content in prompts, rerun compute budget.
Total	**14**	**Hard**